

Coherence-Gated Learning: State-Dependent Plasticity for Continual Learning

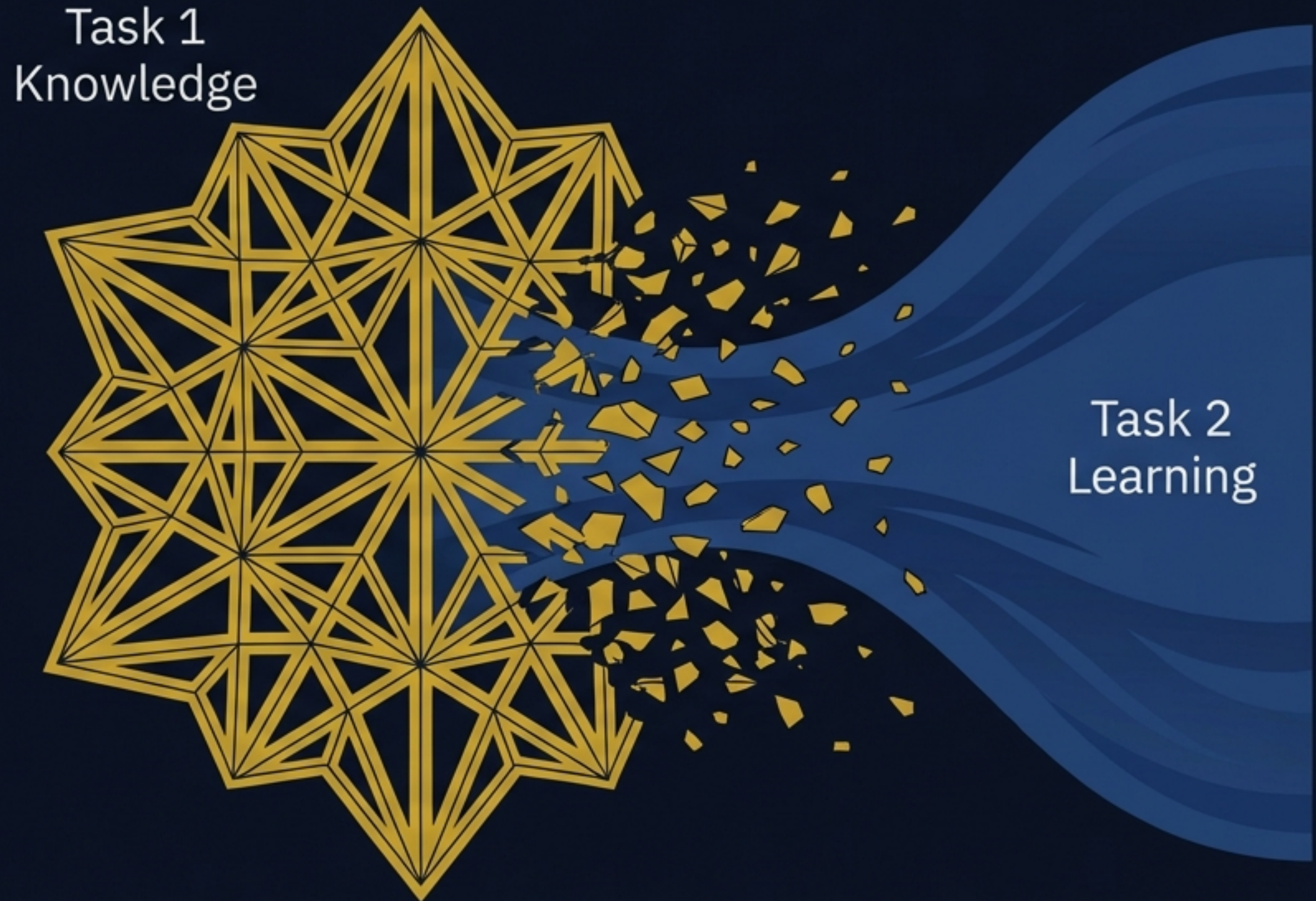
Hadi Servat

FRC Labs | Digid Inc., Barrie, Ontario, Canada

Preprint | December 2024 | DOI: [10.5281/zenodo.18091126](https://doi.org/10.5281/zenodo.18091126)

The Stability-Plasticity Dilemma is Unresolved

Key Statement: Neural networks trained sequentially suffer from catastrophic forgetting, where new learning overwrites previously acquired knowledge. This fundamental tension between retaining past knowledge (stability) and acquiring new knowledge (plasticity) has driven decades of continual learning research. Standard gradient-based learning inherently favors plasticity, leading to a decay of performance on older tasks as new ones are introduced.



Prevailing Methods Treat the Dilemma as a Static Trade-off

Existing solutions attempt to balance stability and plasticity using fixed rules and hyperparameters. They fall into three main categories:

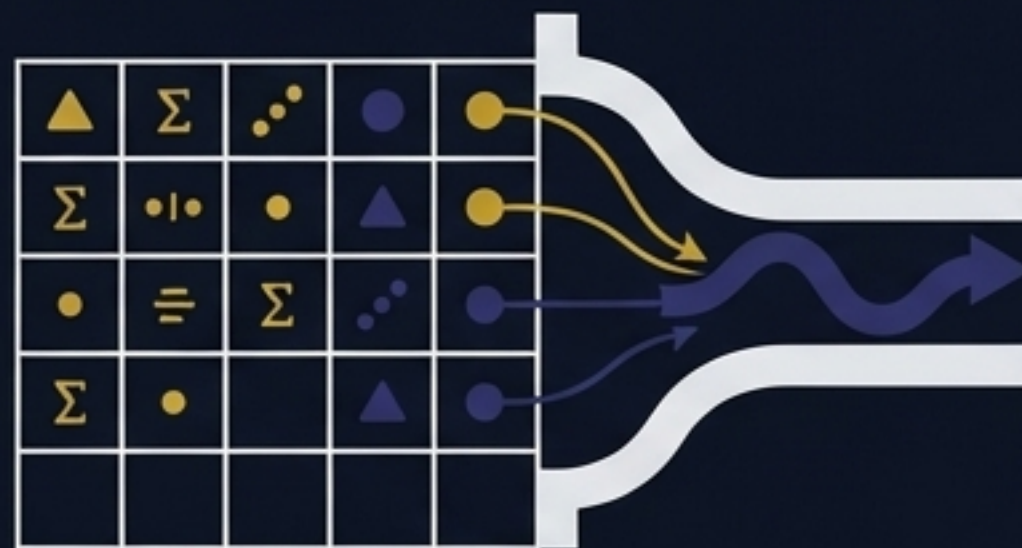
1. Regularization (e.g., EWC, SI)

Penalize changes to weights deemed important for past tasks. This is a static constraint on plasticity.



2. Replay (e.g., GEM, ER)

Maintain a memory buffer of past examples, mixing old and new data. The replay strategy is typically fixed.



3. Architectural

Allocate dedicated parameters per task, physically separating knowledge. This is a rigid, pre-defined separation.



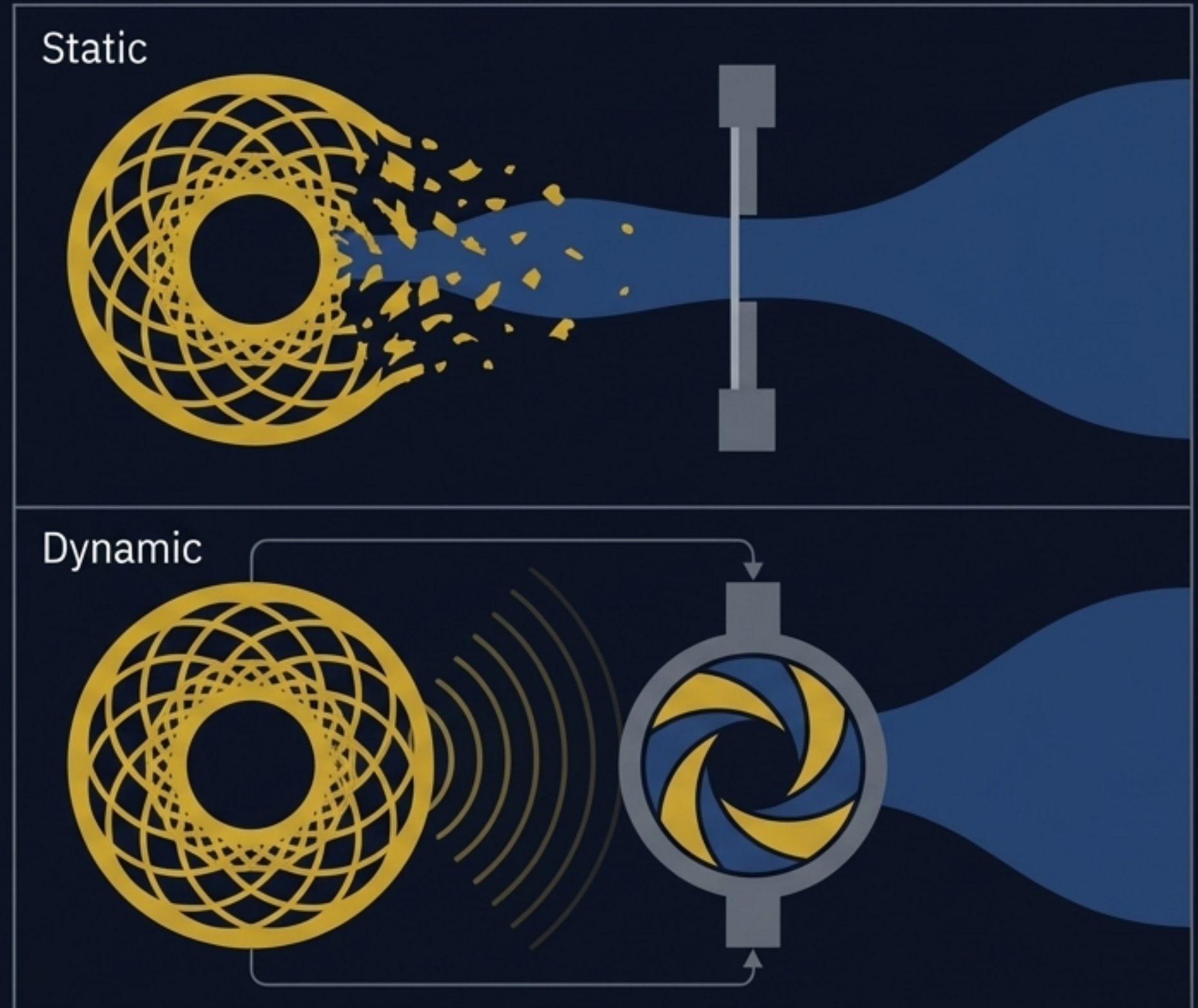
All these approaches impose a *static* balance.
The system itself has no dynamic control over its own plasticity.

The Paradigm Shift: Plasticity Should Be State-Dependent

Core Insight: In biological systems, learning is not constant. It is dynamically modulated by internal states like arousal and attention. Neuromodulatory signals gate when learning occurs.

Proposed Principle: We propose that a system's plasticity should be gated by its own internal coherence dynamics.

- When consolidating knowledge (high coherence), plasticity should be reduced to protect stability.
- When exploring new information (low coherence), plasticity should be increased to facilitate learning.



Step 1: Quantifying System State with Coherence

Concept: We define Coherence, $C(t)$, as the negative entropy of the model's output distribution. It is a direct measure of predictive confidence.

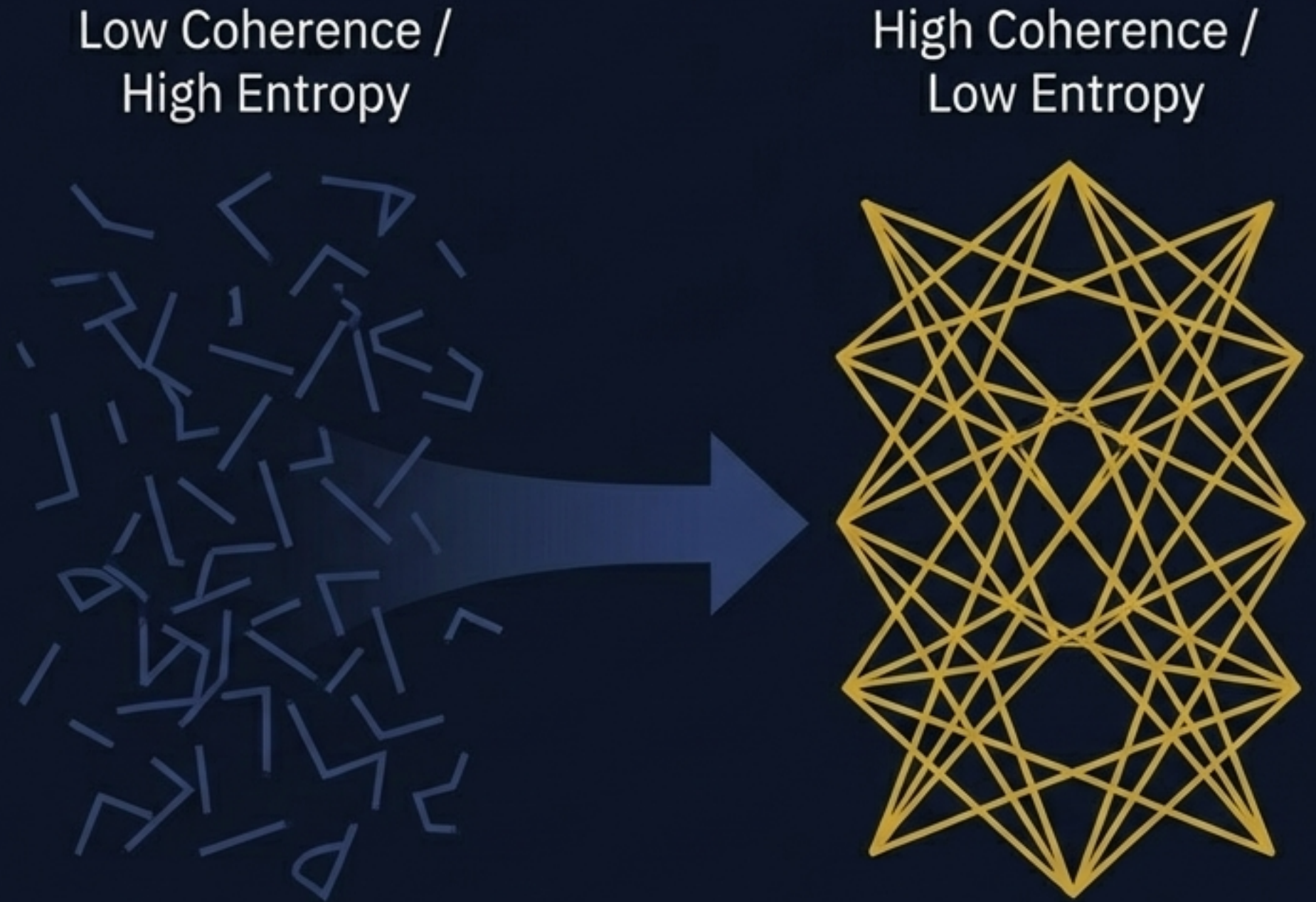
Formula:

$$C(t) = -H(p) = \sum_i p_i \log p_i$$

(Where p_i is the probability of class i)

Interpretation:

- High Coherence: Low entropy, peaked distribution. The model is confident and consolidated.
- Low Coherence: High entropy, uniform distribution. The model is uncertain and exploring.



Step 2: Measuring Coherence Dynamics with Alpha Drift

Concept:

A static coherence value is insufficient. The critical signal is its rate of change, which we term Alpha Drift, $\alpha(t)$.

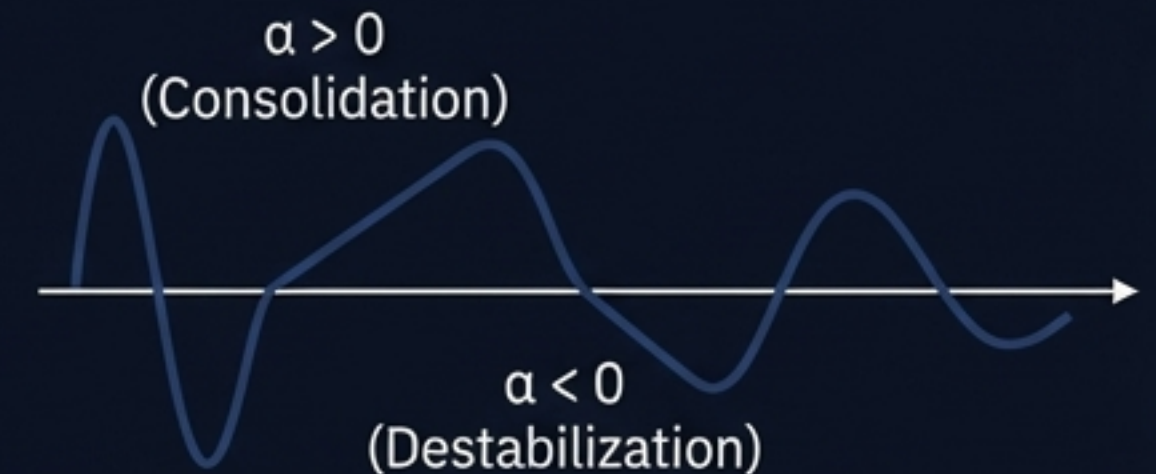
Formula:

$$\alpha(t) = \bar{C}(t) - \bar{C}(t - 1)$$

(Where $\bar{C}$ is an exponential moving average of coherence with $\tau = 0.9$)

Interpretation:

- Positive Alpha ($\alpha > 0$): Coherence is increasing. The system is consolidating knowledge.
- Negative Alpha ($\alpha < 0$): Coherence is decreasing. The system is being destabilized by new information.



Technical Note:

Raw alpha values are normalized using a rolling z-score to ensure scale-invariance across tasks.

$$\tilde{\alpha}(t) = (\alpha(t) - \mu_a) / (\sigma_a + \epsilon)$$

Step 3: Operationalizing Dynamics with a Learning Gate

Concept: The normalized alpha drift, $\tilde{\alpha}(t)$, directly controls a learning gate, $g(t)$, which determines whether the model should update its weights.

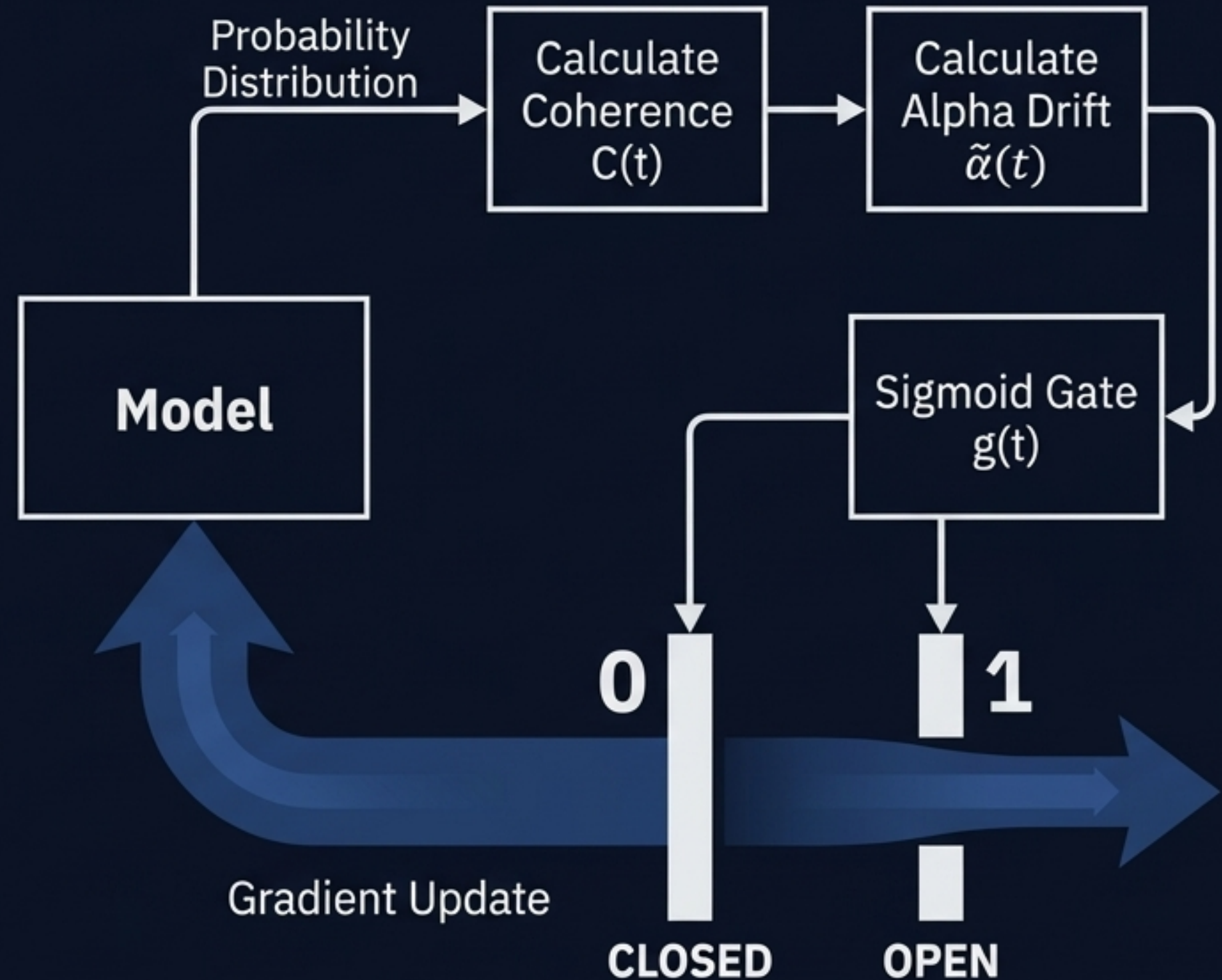
Formula:

$$g(t) = \sigma(k \cdot \tilde{\alpha}(t))$$

(Where σ is the sigmoid function and k controls sensitivity, default $k=2$)

Mechanism (Hard Gating):

Gradient updates are skipped entirely if $g(t) < 0.5$. This provides genuine compute savings. The network actively decides when not to learn.



CGL Outperforms the Baseline on All Metrics While Reducing Compute

Context: Results from the **Split-MNIST** benchmark (5 sequential binary classification tasks).

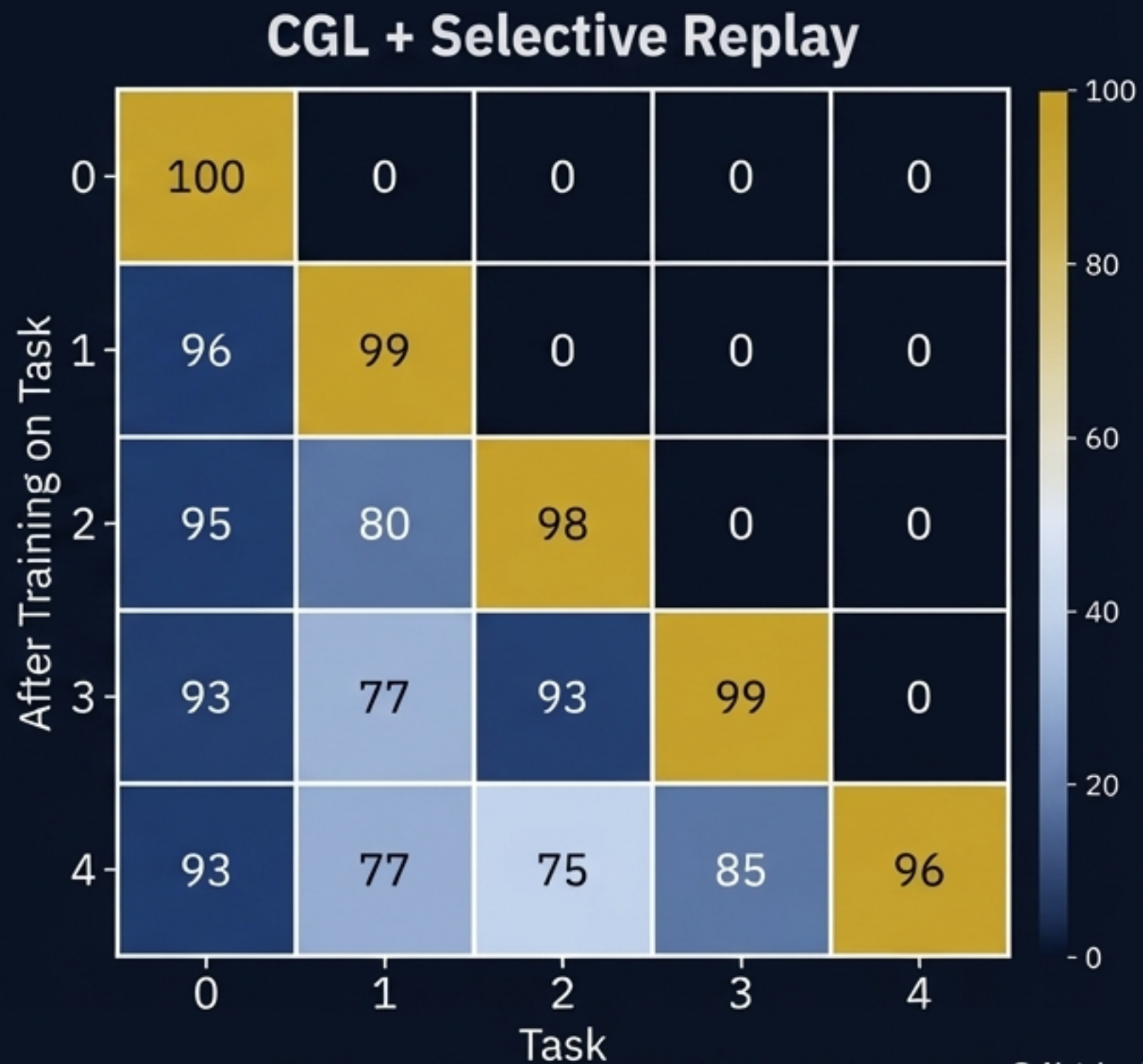
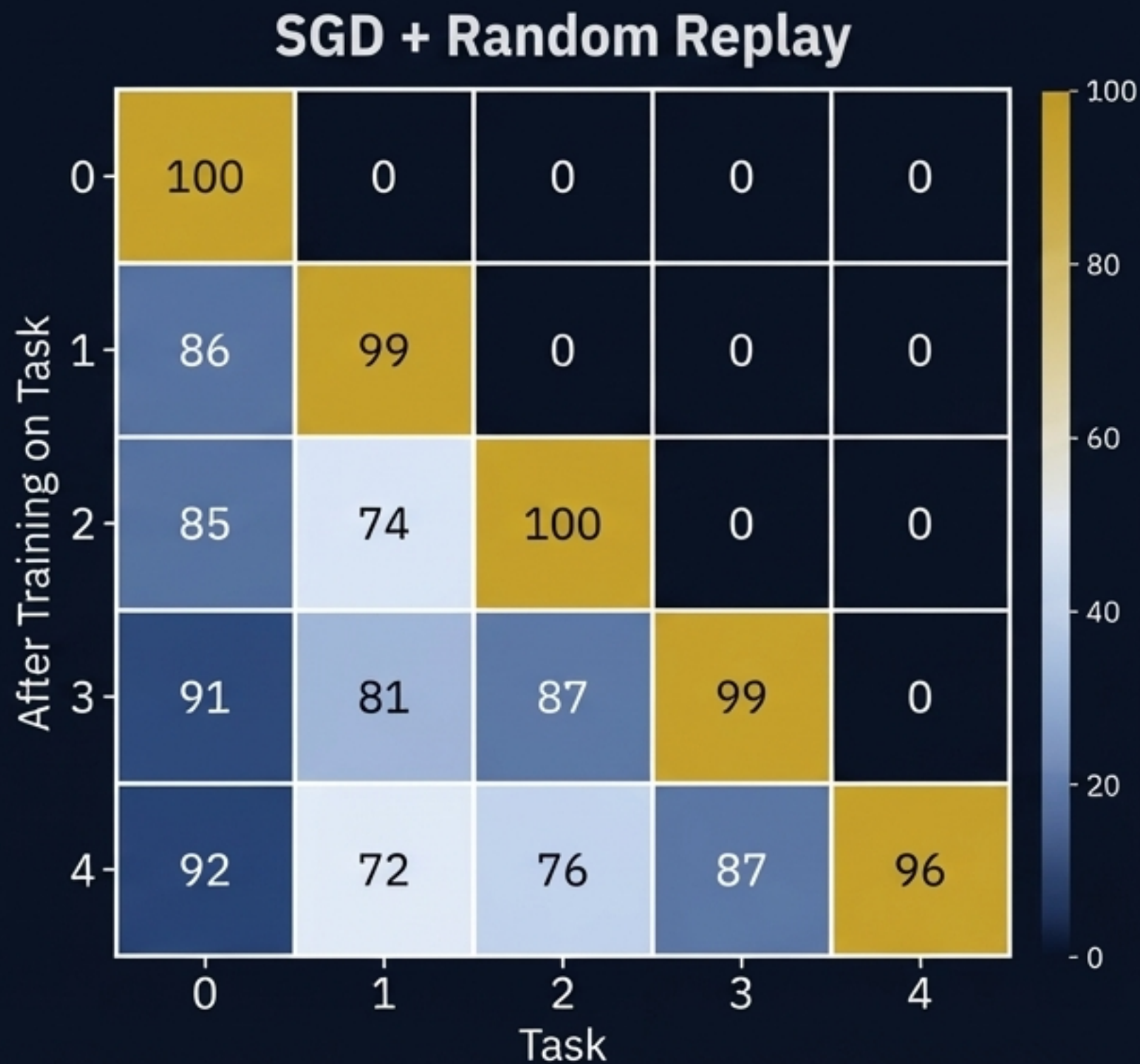
Method	Avg. Accuracy ↑	Backward Transfer ↑	Efficiency
SGD + Random Replay	84.68%	-17.79%	100%
CGL + Selective Replay	85.24%	-16.73%	66.8%

CGL achieves superior **performance** while skipping **33.2%** of all gradient updates.

Note: 'Selective Replay' prioritizes samples from periods of high absolute alpha drift, mimicking biological memory consolidation.

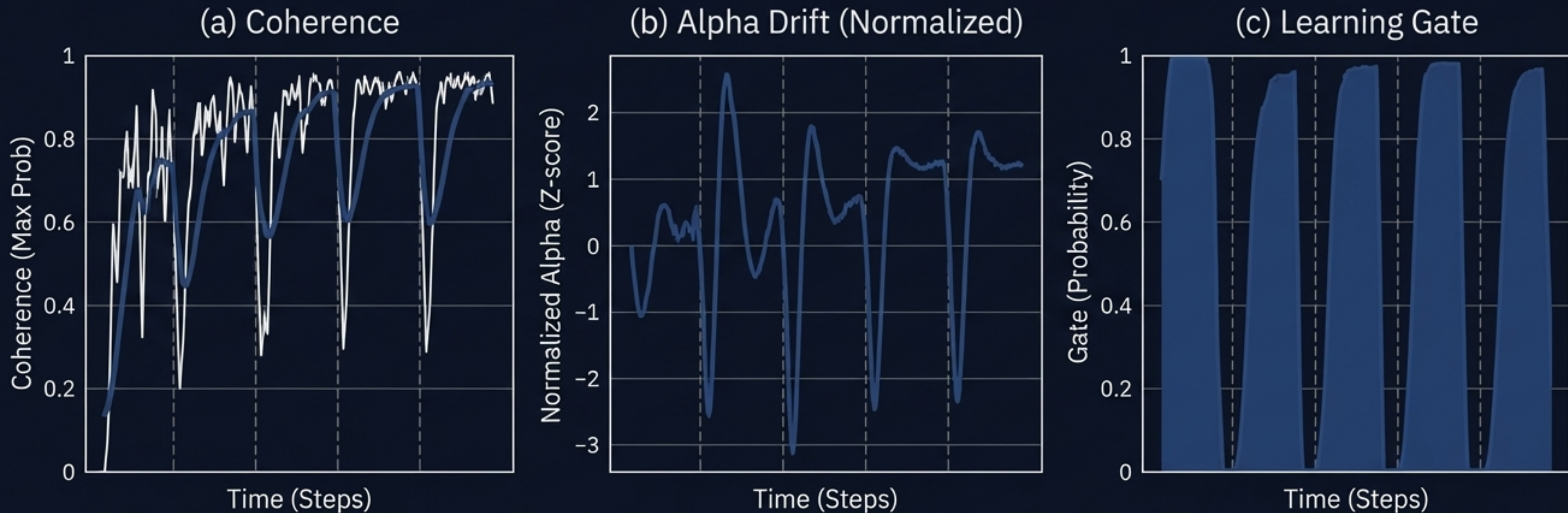
Visualizing Reduced Forgetting in Accuracy Matrices

Description: Accuracy on task j (column) after training on task i (row) for the Split-MNIST benchmark. Higher values in the lower-left quadrant indicate better knowledge retention.



The Internal Dynamics of Coherence-Gated Learning

These plots visualize the internal state of the CGL model during sequential training on the 5 Split-MNIST tasks.



Key Insight: The gate clearly closes (value drops to 0) as the network consolidates on each task and opens (value rises to 1) at the introduction of a new task.

Hard Gating Provides Significant Gains Even Without Replay

To isolate the effect of the gating mechanism, we compare performance without any experience replay buffer.

Method	Avg. Accuracy ↑	Forgetting ↓	Updates Used
Vanilla SGD	63.9%	34.5%	100%
EWC	64.3%	34.2%	100%
CGL-Hard	67.3%	30.1%	41%

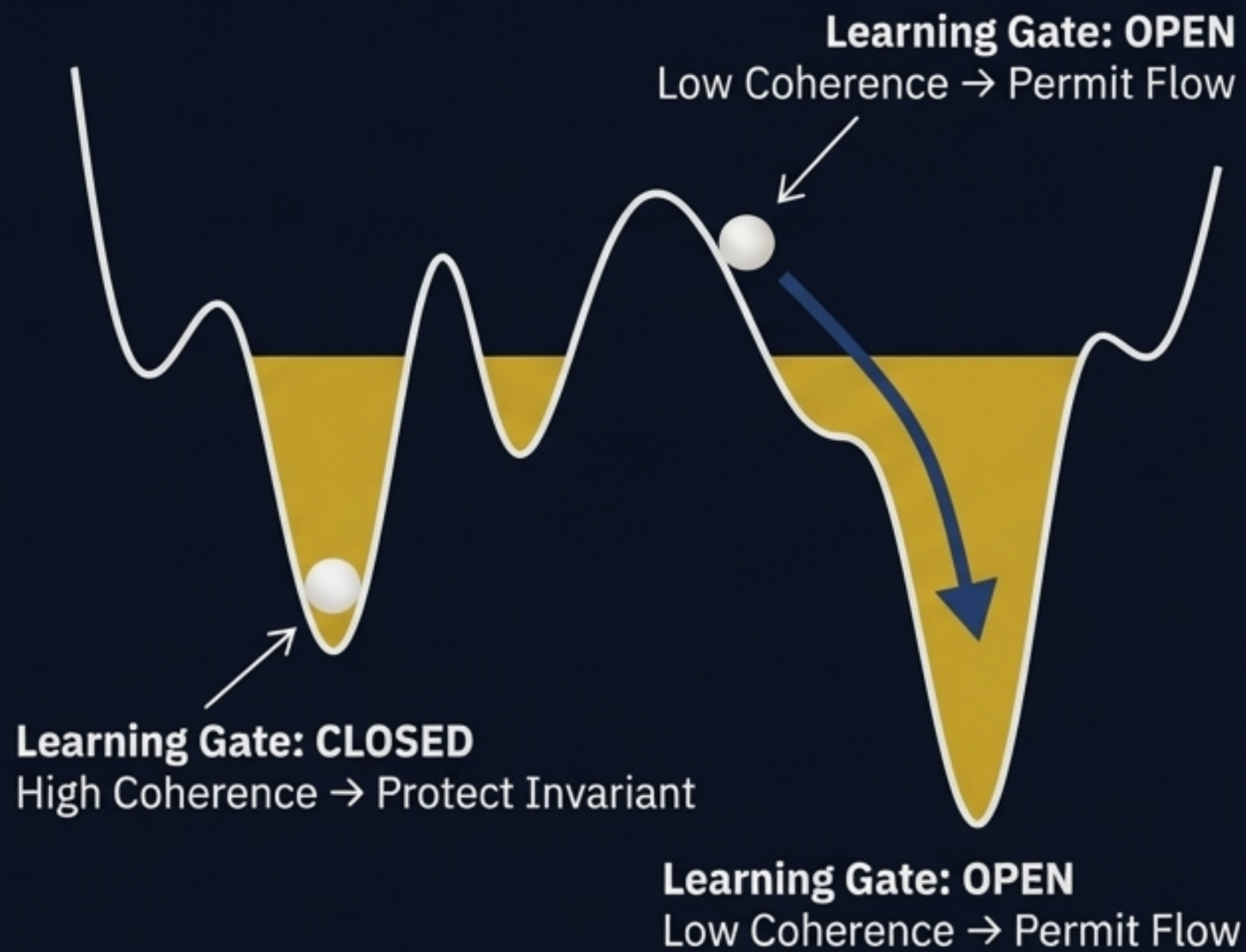
Key Finding: Without replay, CGL-Hard achieves **~5% higher accuracy** and **13% lower forgetting** than SGD, while using **59% fewer gradient updates**. This demonstrates the raw power of state-dependent learning.

CGL Works by Protecting Stable Attractor States

Coherence-based gating intuitively works when the network is in a stable attractor (representing learned knowledge) versus when it is transitioning between attractors.

- **During Consolidation (High Coherence, $\alpha > 0$):** The network is in a stable attractor. Gradients are small or noisy. Skipping updates protects this consolidated state from corruption.

- **During Exploration (Low Coherence, $\alpha < 0$):** The network is transitioning between states. Gradients are informative and point toward a new solution. Updates must proceed to acquire new knowledge.



A Computational Analogue to Biological Neuromodulation

Principle: CGL operationalizes the biological principle that *when* to learn should be separated from *what* to learn.

Biological System	CGL Analog
Neuromodulators (Dopamine, NE)	Learning gate $g(t)$
Arousal / Attentional State	Normalized alpha drift $\tilde{\alpha}(t)$
Sleep / Consolidation	High Coherence (Gate Closed)
Active Learning / Surprise	Low Coherence (Gate Open)

Limitations and Future Directions

Current Limitations

- **Benchmarks:** Results are primarily on Split-MNIST; evaluation on more complex benchmarks (e.g., CIFAR-100) is ongoing.
- **Gain Magnitude:** The absolute improvements in accuracy (0.56%) and BWT (1.06%) are consistent but modest.
- **Hyperparameter Sensitivity:** The EMA decay (τ) and gate sensitivity (k) may require tuning for different domains.

Future Work

- **Online Task Boundary Detection:** The current method assumes known task boundaries. Future work will explore detecting these shifts automatically from coherence dynamics.
- **Integration with Other Methods:** Exploring synergies between CGL and advanced regularization or replay techniques.

The Core Insight: Learning Should be Gated by Internal State

Summary: Coherence-Gated Learning demonstrates that *when* to learn should be determined by internal state dynamics, not just external error signals. By monitoring the system's own confidence and gating plasticity based on its rate of change, CGL achieves better accuracy, reduced forgetting, and significant compute savings.

A network that knows when it is stable should protect that stability. A network that knows when it is uncertain should embrace plasticity. CGL provides a principled, biologically-motivated mechanism for this state-dependent learning.